Studying the Leprosy Crisis in India

By Zahabiya Murtuza Sakerwala

Acknowledgement

I would like to thank Mr Anthony Samy, the Chief Executive of ALERT INDIA for his invaluable insights into the leprosy challenges in India.
Since our first meeting in 2024, our discussions have deepened my understanding of the struggles faced by leprosy patients today. Seeing his dedication to supporting those in need has been the source of inspiration for this research paper.
Following our initial conversation, Mr Anthony was kind enough to send me a hefty number of books and pamphlets containing literature developed by ALERT INDIA over the years.
I am pleased to acknowledge that most of this material has been featured in this research paper.

Abstract

In December of 2005, the WHO announced that Leprosy has been eliminated from India. The definition of elimination - <1/10,000 cases of leprosy infected patients. This announcement brought a wave of optimism among Indians, to a point where the term elimination was often confused with eradication. This statement resulted in the reduction of active detection of Leprosy cases, reduced effort to educate people about Leprosy and reduced resource allocation for Leprosy. Since then Leprosy workers have argued the premature declaration of “elimination” and have challenged that the number of actual cases of Leprosy is far more than the reported ones, attributing it to the biased goal of validating the 2005 WHO announcement. Till today, India is the largest contributor of Leprosy in the world. The nature of leprosy and the implications of this announcement are still felt today. This research paper takes into account articles and research papers from Leprosy Programme Managers in India and healthcare professionals to analyse the Leprosy crisis in India.

Introduction

Leprosy is thought to be one of the oldest diseases known to man, with its roots in India dating back to 2000 BC. While the debate has been around if Leprosy originated from Africa or India, the problem at hand is much worse. India has the highest number of leprosy cases in the world, followed by Brazil and Indonesia. During its earlier times, India has made substantial progress in the treatment of leprosy. The introduction of MDT in 1983 (multi drug therapy) fueled the treatment and played an active role in reducing the number of leprosy cases. However this success would soon come crashing down in 2005, when the WHO declared that Leprosy was eradicated in india. This statement was far from true, premature at best. The repercussions of this statement are still felt by many vulnerable Indians today. In this research paper, we will delve into understanding leprosy, the controversial WHO declaration and analysing the damming effects it has caused Indians as a consequence.

What is Leprosy?

Leprosy also known as Hansen disease is a chronic granulomatous infection generally caused by *Mycobacterium leprae* and *Mycobacterium lepromatosis, meaning* that it is a long lasting infection that causes lumps and swelling in the body caused by the two types of bacteria, respectively.
It is an Acid Fast Bacillus (AFB) and an obligatory intracellular organism, meaning that it is the type of bacteria that is not easily destroyed by acids, and can only survive and grown in the cells of a living organism, ie - it cannot be cultured in an artificial medium. The bacteria does not produce any toxins, and hence the affected person feels no discomfort. Only bacteria that invades the peripheral nerves causes neuropathy.

How is Leprosy detected?

According to the WHO, leprosy manifests generally through skin lesions and peripheral nerve involvement. Leprosy is diagnosed by finding at least one of the following signs - 1. Loss of sensation in a definitive pale or reddish skin patch. 2. Nerve near the skin becomes

thick or swollen resulting in numbness or weakness in the muscles controlled by the nerve.
3. A small skin sample tested, finds presence of the bacteria.

Based on this, leprosy can be divided into 2 types for treatment reasons

1. PB - paucibacillary, meaning a leprosy case where the patient has 1-5 skin lesions but no detectable bacteria in the skin smear test.
2. MB - Multibacillary, meaning a case of leprosy where the patient has over 5 skin lesions, or with nerve damage (can manifest without skin patches), or detectable presence of bacteria in the skin smear test, regardless of the number of patches.

How does one get Leprosy?

According to the Third Edition of *'We can detect, treat and cure leprosy - A Guide for Public Health Doctors' published by ALERT INDIA,* untreated individuals hosting an infection of the *Mycobacterium leprae* and *Mycobacterium lepromatosis* are the main source of spread of Leprosy. Leprosy can spread when an infected individual releases a large number of bacilli (Bacilli are rod-shaped bacteria) from the upper respiratory tract, ie - through sneezing or coughing, making the cause of spread airborne by droplet infection. The disease does not spread through casual contact like sitting next to an infected person, shaking hands, hugging, etc. Additionally, the patient stops transmitting the disease once the treatment has begun. The leprosy bacilli multiply very slowly. One bacillus can take 15-20 days to divide into two. In an immunocompromised individual, the body's natural immunity does not trigger a strong response to the bacilli. This is due to the low antigenic property of the bacilli, ie - the bacilli does not strongly trigger the body's immune system, making it harder for the body to recognize and respond to the threat and fight off the infection.

How can Leprosy be treated?

The silver lining is that leprosy is a treatable disease. The course of treatment includes a mix of three medicines - Rifampicin, Clofazimine and Dapsone also referred to as the Multi drug therapy MDT. The course of treatment for *Paucibacillary Leprosy (PB)* is 6 months, with the use of Rifampicin and Dapsone. The treatment for *Multibacillary Leprosy (MB)* is 12 months, with the use of Rifampicin, Clofazimine and Dapsone. MDT kills the pathogen and cures the patient. Early detection and prompt treatment can prevent disability in infected individuals, however for individuals where the treatment was delayed and have experienced nerve damage, living and coping with the disability becomes a lifelong battle.

The state of Leprosy in India

According to *Imran Ahmed Khan and MD. Abu Bashar*, Leprosy is one of the oldest diseases known to mankind. It is a chronic infection burdened with the title of an NTD or Neglected Tropical Disease. Leprosy can infect individuals of all ages, infants to elderly. Through the years, work in combating leprosy has proven effective, with the invention of MDT, the spread of leprosy has been restricted drastically. However Leprosy remains to be a health concern in India.

A point to note is that the definition of elimination of Leprosy, ie - identifying that Leprosy elimination is no longer a public health concern, is a point where less that 1 person for every

10,000 is a leprosy patient. Most countries achieved this target with a scattered number of cases here and there by the year 2000. In India, this target was achieved by 2005.

Regardless of this valuable achievement, India continues to house 52% of the world leprosy population. Experts believe that more cases are being reported due to the testing initiatives taken by organizations like LCDC (Leprosy Case Detection Campaign) and SLAC (SPARSH Leprosy Awareness Campaign).

Leprosy is endemic in several states and union territories in India. With an annual case detection of 4.56 per 10000 populations. Chhattisgarh (state) and Dadra and Nagar Haveli (Union Territory) have not achieved elimination status yet.

In a report by *P Narasimha Rao and Sujai Suneetha*, In 2007 new cases of Leprosy detected in India were 137,685, nine years later in 2016, the number remained almost unchanged at 135,485. The increase in detection of new cases was all thanks to the NLEP's strategy The Leprosy Case Detection Campaign (LCDC), an innovative initiative, that identified 34,000 new cases in 2016 from high-risk areas, contributing to 25% of the total new cases reported that year. Out of the total number of new cases reported, 50% were multibacillary, with a child rate of 8.7%. The LCDC's efforts also showed an increase in the number of districts where the leprosy detection rate was >1/10,00 cases, ie - not in alignment with the definition of leprosy elimination.

As is seen here, if the number of Leprosy cases are rising, the question arises. What was the reason behind the hasty decision to announce Leprosy elimination? And what is Leprosy Elimination?

Analyzing the WHO definition of Leprosy elimination

In an article by *Apoorva Mandavilli (Science Journalist, The New York Times)* the WHO had long made Leprosy elimination one of its major goals. In 1991 the organisation announced that by the year 2005, leprosy would be eliminated as a public health problem. Ie <1/10,000 cases.

The odds were long but this goal had been deferred before. The repercussions of this? Dr. Anil Kumar who has led the country's leprosy programme since 2015 said that oftentimes program managers made less of an effort to find the infected cases, enjoying the hoax of detecting lesser cases.

Before we look at the possible hidden agenda of the WHO to declare leprosy elimination, let's understand the difference between 'elimination' and 'eradication'.

Erradican of leprosy would indicate that leprosy has completely vanished from the face of the earth, zero cases of leprosy through a consequence of deliberate testing and treatment of infected individuals.

Elimination in the case of leprosy means <1/10,000 infected cases.

According to the WHO, the reason for setting the goal of elimination was due to the success of MDT in treating and controlling the spread of leprosy. To the public, it sounds like a reasonable goal to have. Maybe even sounds impressive. Eliminating one of the oldest diseases on earth is definitely an achievement for the WHO.

WHO's possible hidden agenda to declare Leprosy elimination

To further understand this, let's break it down into 3 parts

1. *Resource allocation*

In a report by *Seong Hye Jung et al*, one potential consequence of declaring elimination meant shifting resources from leprosy to other areas of interest. In the case of leprosy, the disease doesn't cause death. Instead it causes disability. Meaning an individual born perfectly healthy who during the course of their life may have contracted leprosy and received treatment after the disease has done irreversible damage, is now burdened with disability for the rest of their life. Meaning that when allocating resources for Leprosy, the cost of treatment (MDT) is not the only thing to factor in. Resources also need to be allocated for people who are disabled as a result of Leprosy.

2. *Integration challenges*

Integrating a specialised disease like Leprosy into the general healthcare systems is more complex than it seems.
Firstly, Leprosy manifests in a variety of ways, in some cases with an evident skin patch (which may or may not detect leprosy after a skin test), nerve damage (loss of sensation, which again may or may not be an indication of leprosy), etc. To add to the confusion, leprosy can remain dormant in an infected individual for up to 20 years.

Secondly, ensuring regular MDT supply can be tricky. The course for leprosy treatment ranges from 6 months to 12 months. Keeping a track on the number of patients receiving their prescribed doses of MDT becomes a challenge. Making the integration of leprosy into the general healthcare system a tedious challenge.

Lastly, reaching the underserved areas is difficult. By integrating leprosy into general healthcare systems, means making an active effort to reach underserved areas. Failure to do so can cause upheaval among people claiming unequal distribution of healthcare. This problem is more severe in the case of Leprosy since the people at risk of infection are immunocompromised individuals ie - higher likelihood of the underserved population being a victim of Leprosy.

3. Convenience

The third possible agenda is convenience. Let's come back to the WHO's choice of words. Their goal was to announce that Leprosy has been eliminated, not eradicated. The two words, when looked at closely, are poles apart, but to the general public can often be misunderstood for the latter. An article by the *Association of Healthcare Journalists* says that sometimes even journalists confuse the 2 terms.

Dr Diana Lockwood, a Leprosy expert at the London School of Tropical Medicine, UK says “for them, elimination means no cases” The general perception of “no cases” means lesser demand to tackle leprosy problems, lesser demand conveniently translates to lesser funds and resources allocated to curing and preventing Leprosy.

Consequences of declaring Leprosy elimination in India

Leprosy is the least infectious of all communicable diseases. Studies show that leprosy can remain dormant in an infected patient for up to 20 years, that means up to 20 years of spreading Leprosy without detection. W.H. Jopling, a consultant Leprologist in London says “Danger to others, if any, arises not from Leprosy patients under treatment but from undiagnosed cases”.

In an article published in the *7th Edition of FOCUS* - an initiative by ALERT INDIA to raise awareness on leprosy, the damming effects of premature declaration of elimination of the Leprosy, an article by *Binod K Patro et all*, International Task Force for Disease Eradication established criterias to support the potential eradication of diseases, building on the success of the eradication of smallpox, they evaluated over 90 diseases. They concluded that only six were possible to eradicate. Leprosy did not make it to the list.

The shocking part about this finding is that India named its National program NLEP ie - National Leprosy Eradication Programme. This shows that the social and political opinions have taken an upper hand in this decision.

The term reflected by the WHA - World Health Assembly’s resolutions ‘Elimination of a disease as a public health problem’ seems to create more of a confusion.

When the Leprosy agenda was announced for 2005, Leprosy elimination efforts were targeted to controlling the disease, rather than controlling the infection. It appeared that most people, including health care workers did not understand the concept of elimination, thinking that it instead meant absence of cases. Though the target of Leprosy elimination was reached in 2006, India till date remains the largest contributor of leprosy in the world.

The biological and epidemiological evidence suggests that leprosy cannot be eliminated simply through MDT, The reason is that scientists still do not fully understand how the bacteria that cause leprosy survive and spread (its microbiology is not fully elucidated).

While the elimination goal spread optimism through the country, it also fueled people to work against Leprosy. They stopped identifying leprosy as a threat, in turn discouraging research in this very field, at a time when India needs it most.

The shift in importance from Leprosy doesn’t just affect science and is well beyond numbers. It fuels the stigma leprosy patients already face.

In the *South Sudan Medical Journal, August 2010, Edward Eremugo Luka* classified stigma in to three types

1. Enacted - where an individual faces actual experiences of discrimination.

2. Perceived - stigma that is viewed from the patient's perception. Shame triggered by attributing negative stereotypes to oneself.
3. Self stigma - fear of being discriminated against.

Health care workers that actively take part in the work for Leprosy often highlight that more than the treatment of leprosy, battling the stigma patients face is worse. Anthony Samy, the Chief Executive at ALERT INDIA believes that this stigma is due to the nature of Leprosy. No doubt that for years people have feared Leprosy, with its uncouth manifestation in fables, people tend to isolate leprosy infected patients out of fear that they may contract it. While this is possible, the possibility of contraction is much lesser than diseases like Tuberculosis. Despite that, Leprosy patients are discriminated against way more than Tuberculosis patients. One possible explanation is that the worst outcome of communicable diseases like TB and AIDS is death, whereas leprosy causes only disability. Government and non government organisations valued combating the stigma TB and AIDS patients face justifying it to improving their quality of life, where ironically the “unattractive” outcome of Leprosy caused it to be sidelined.

The effects of the stigma attached to leprosy can have damming effects on the individual. Namely the diagnosis being worse than the actual symptoms. The stigma attached has been seen to result in emotional stress and anxiety which can lead to psychological and psychiatric morbidity and can in turn cause a decreased quality of life. Due to the notion that Leprosy is fast spreading, oftentimes victims are abandoned, decreasing their status in the community and in unfortunate instances leaving them on the streets.

The effects on the individual can translate in 2 major areas

1. *Community interaction*

Their diagnosis prevents them from interacting with their community. They face discrimination when getting married, added difficulty in gaining employment, adversely affecting their relationships, friendships and community participation.

2. *Public health programmes*

Due to the fear of being discriminated against, potential patients may deny or conceal their symptoms, leading to delayed treatment, running the risk of increasing transmission of the disease.

ALERT INDIA identified these challenges and created a process that tackles both, the education problem and the treatment problem for leprosy. By setting up testing camps in districts and villages, they treat patients that have been infected with Leprosy, while paying heavy attention in educating people about Leprosy, how to detect it and most importantly debunking the myths behind Leprosy ensuring patients have a better quality of life.

The Global Leprosy Strategy 2021 - 2030 WHO

It is refreshing to see that after years of struggle, the WHO collected valuable insights from Leprosy programme Managers technical agencies, public health and Leprosy experts, funding agencies and persons or members of communities directly affected by Leprosy, and introduced the Global Leprosy Strategy 2021 - 2030 Towards Zero Leprosy. This strategy

aims to achieve the goal through integrating country specific Zero Leprosy roadmaps in endemic countries, scaling up Leprosy prevention through active case detection, managing Leprosy and its contribution through preventing new disability and combating the stigma ensuring human rights are respected.
While this effort of the WHO is a response to the continuous work of Leprosy workers, it is important to note that the WHO has not redacted its premature statement that Leprosy has been eliminated in India. While evidence still suggests that some states and union territories in India still fall behind the threshold of 'Leprosy elimination'.

Conclusion

Leprosy is one of the oldest battles still being fought by man. India being the largest army combating it. The Indian government has collaborated with the WHO to implement strategies that can help integrate Leprosy treatment with general healthcare. The Indian government released the NSP - National Strategic Plan, which contains a roadmap for leprosy (2023 - 2027) to achieve the goal of zero transmission of leprosy by 2027.
While this may seem like an optimistic plan for most, using our own tried and tested roadmap that followed the declaration made by WHO in 2005, the scepticism outshines the optimism. How many more cases will go undetected? Will India actively place its resources to detect and treat leprosy or will they opt for the easier path - out of sight, out of mind. The gravity of the disease is often overlooked. Leprosy causes disability, It causes the victim to lose the freedom they were born with, garnished with seclusion, discrimination and a poor quality of life.

Citations

1. Sil, Abheek, and Anupam Das. “History of Leprosy in India: An Overview of Historic and Modern Contributions.” *Clinics in Dermatology*, vol. 40, no. 6, July 2022, pp. 691–99. https://www.sciencedirect.com/science/article/abs/pii/S0738081X22000918?via%3Dihub

2. Katoch, VishwaMohan. “Eradication of Leprosy From India: Reflections on Past, Present &Amp; Future.” The Indian Journal of Medical Research, vol. 159, no. 1, Jan. 2024, p. 1. https://journals.lww.com/ijmr/fulltext/2024/01000/eradication_of_leprosy_from_india__reflections_on.1.aspx

3. Bhandari, Jenish, et al. “Leprosy.” StatPearls - NCBI Bookshelf, 15 Sept. 2023, www.ncbi.nlm.nih.gov/books/NBK559307.

4. World Health Organization: WHO. Leprosy. 24 Jan. 2025, www.who.int/news-room/fact-sheets/detail/leprosy.

5. Bashar, Md. Abu, and Imran Ahmed Khan. “Challenges of Sustaining Leprosy Elimination in India and Prospects of Eradication: Current Status &Amp; Way Forwards.” Indian Journal of Forensic and Community Medicine, vol. 11, no. 1, Apr. 2024, pp. 11–14. https://www.ijfcm.org/article-details/21473

6. Rao, P. Narasimha, and Sujai Suneetha. “Current Situation of Leprosy in India and Its Future Implications.” Indian Dermatology Online Journal, vol. 9, no. 2, Jan. 2018, p. 83. https://journals.lww.com/idoj/fulltext/2018/09020/current_situation_of_leprosy_in_india_and_its.1.aspx

7. Union Minister of Health and Family Welfare, Dr. Mansukh Mandaviya Virtually Addresses National Anti-Leprosy Day. pib.gov.in/PressReleasePage.aspx?PRID=1894687.

8. Control of Neglected Tropical Diseases (NTD). Towards Zero Leprosy. Global Leprosy (Hansen’s Disease) Strategy 2021–2030. 15 Apr. 2021, www.who.int/publications/i/item/9789290228509.

9. Jung, Seong Hye, et al. “Patients Help Other Patients: Qualitative Study on a Longstanding Community Cooperative to Tackle Leprosy in India.” PLoS Neglected Tropical Diseases, vol. 14, no. 1, Jan. 2020, p. e0008016. https://journals.plos.org/plosntds/article?id=10.1371/journal.pntd.0008016

10. Nsagha, Ds, et al. “Elimination of Leprosy as a Public Health Problem by 2000 AD: An Epidemiological Perspective.” Pan African Medical Journal, vol. 9, no. 1, Oct. 2011, https://www.ajol.info/index.php/pamj/article/view/71176

11. Sengupta, Utpal. “Elimination of Leprosy in India: An Analysis.” Indian Journal of Dermatology Venereology and Leprology, vol. 84, no. 2, Jan. 2018, p. 131. https://ijdvl.com/elimination-of-leprosy-in-india-an-analysis/

12. Association of Health Care Journalists. “Disease Elimination Vs. Eradication | Association of Health Care Journalists.” Association of Health Care Journalists, 9 Nov. 2023, healthjournalism.org/glossary-terms/disease-elimination-vs-eradication.

13. ALERT - INDIA. (2013). WE CAN DETECT TREAT CURE LEPROSY - a guide for public health doctors (3rd ed.).

14. ALERT - INDIA. (2015). TASK TODAY (6th ed.).

15. ALERT - INDIA. (2020). FOCUS (7th ed.).